
Drift–Utility Dissociation in Time-Series Foundation Model Fine-Tuning: Methodology and a Moirai Case Study

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Abstract

1 We contribute a **reusable diagnostic methodology** for assessing representa-
2 tion drift during fine-tuning of time-series foundation models: a value-gate
3 screening criterion, controlled-comparison design with frozen-encoder/random-
4 init/layer-wise-unfreeze controls, task-orthogonal probe protocol, LoRA-Large
5 LR recipe, uni2ts library patch, and CUDA-deterministic reproducibility proto-
6 col. We demonstrate the methodology on a **Moirai-ETT case study** (the one
7 backbone×dataset family where the value-gate passes). The load-bearing empiri-
8 cal signals are: (1) the **task-orthogonal probe asymmetry** ($\Delta R^2 < 0$ on positive-
9 floor probes, $\Delta R^2 > 0$ on the trained head; 10/10 seeds on ETTh2 and ETTm2,
10 where ETTm2 is a gate-failing cross-domain comparison confirming the mech-
11 anism generalises), operationalising “task-specific shift”; and (2) **layer-wise lo-**
12 **calisation** (top-3 layers suffice for $\Delta R^2 > 0$ but lower layers required for task re-
13 covery; 10/10 seeds). Fine-tuning imposes geometric cost (CKA 0.949→0.52 as
14 $n:500 \rightarrow 10k$) with protocol-dependent task outcome—harmful at $n \leq 1k$ (+14%
15 forgetting), improving at $n=10k$ under early-stopped checkpointing. Five non-
16 Moirai backbones gate-fail on all nine ETT cells tested; Moirai itself gate-fails on
17 Traffic, Exchange, and Electricity, narrowing the gate-pass to ETTh2 and confirm-
18 ing that corpus composition drives coverage (§5.1). **Practitioner takeaway:** Use
19 LoRA with $10\times$ LR reduction on larger Moirai variants before escalating rank;
20 apply the value-gate before investing in drift diagnosis.

21 1 Introduction

22 Moirai Woo et al. [2024] is a foundation time-series model trained on billions of time points that
23 achieves strong zero-shot performance on forecasting benchmarks. A natural next step is to fine-tune
24 it on domain-specific tasks: anomaly detection, classification, or short-horizon forecasting with lim-
25 ited labelled data. We present a Moirai case study on ETT forecasting benchmarks that characterises
26 what happens to pre-trained representations during fine-tuning.

27 **Representation drift during fine-tuning.** We contribute a systematic *diagnostic methodology* for
28 characterising representation drift—progressive overwriting of pre-trained features—during fine-
29 tuning of time-series foundation models, and demonstrate it on Moirai (Small 13.8M, Base 91M,
30 Large 311M) as the single backbone family where pre-trained features pass our value-gate on ETT
31 benchmarks. Matched-protocol attempts to replicate on three non-Moirai backbones (Chronos-T5-
32 Small Ansari et al. [2024], TimesFM-2.5-200M Das et al. [2024], MOMENT-1-base Goswami et al.
33 [2024]) gate-fail on all nine ETT×backbone cells tested (7 original + 2 new Chronos capacity cells);
34 we report this as a substantive finding about backbone-specific ETT pre-training coverage (§4, Ap-
35 pendix A). Catastrophic forgetting during fine-tuning is well-studied in continual learning Kirk-

patrick et al. [2017], Li and Hoiem [2018] and NLP Luo et al. [2023]; our contribution is reusable diagnostic methodology, demonstrated on a well-controlled Moirai case study. We use “catastrophic forgetting” to mean degraded performance on distributions the model originally handled well—consistent with continual learning usage where the “previous task” is the pre-training objective.

Distinguishing drift from overfitting. Representation drift degrades zero-shot performance while training loss decreases (not overfitting); the frozen-encoder control (CKA=1.000, same data/epochs) isolates encoder weight updates as the cause.

On ETTh2 (gate-pass: Moirai outperforms Linear by 28–45%, $R^2_{\text{task}}(\text{PT})=+0.279$), fine-tuning degrades MSE by 10–16% at $n=500$ and resolves by $n=2,000$. ETTh1 and ETTm2 gate-fail at the standard evaluation protocol and are included as cross-domain comparisons (Table 2). On IoT (sub-random ZS AUC), CKA drops to 0.31–0.45 but task performance improves from a near-null baseline: the drift is irrelevant because there were no valuable features to forget.

Why this matters. Time-series drift directly harms distributional predictions for anomaly scoring and probabilistic forecasting. Consequences depend on pre-training value: harmful on ETT in low-data regime ($n \leq 1\text{k}$), benign at large n once replacements are learned. *Counterfactual without the value-gate:* Chronos-T5-Small on ETTh2 gives ZS MSE=0.304 vs. Linear 0.213 (gate-fail: ZS underperforms by 43%). A practitioner who skipped screening would fine-tune Chronos, observe CKA dropping, diagnose “drift,” and apply LoRA/EWC to preserve features that had *no advantage over Linear*—wasted effort. The value-gate prevents this: drift diagnosis is meaningful only when pre-trained features demonstrably outperform a trivial baseline.

Scope. The empirical case study is narrow by design: one gate-passing backbone (Moirai) on one clean setting (ETTh2, $n=500\text{--}10\text{k}$). We do not claim a universal phenomenon; we claim reusable methodology and one well-controlled case study illustrating its application. The methodology (value-gate, controls, probes, protocol) transfers to any backbone; the specific numeric findings require replication.

Experimental approach. We deploy the diagnostic methodology across three Moirai sizes (Small/Base/Large) on ETT forecasting Zhou et al. [2021] (ETTh2 gate-pass; ETTh1/ETTM2 cross-domain) and IoT anomaly detection (negative control; §5). Forgetting sign at $n=10\text{k}$ is protocol-dependent (−5.3% ES vs. +7.5% final-epoch); we lead on the more robust $\Delta R^2 > 0$ majority.

Contributions. Methodological contributions are offered as reusable infrastructure; empirical findings are Moirai-ETT specific and require replication.

1. **Diagnostic methodology (backbone-agnostic).** Value-gate screening criterion (ZS >20% vs. Linear); controlled-comparison design (frozen-encoder, random-init, layer-wise unfreeze controls); LoRA-Large $10\times$ LR rescue recipe (cross-dataset validated); uni2ts PackedStdScaler patch; and CUDA-deterministic per-seed protocol (Appendix H).
2. **Empirical validation (Moirai-ETT case study).** Task-orthogonal probe asymmetry ($\Delta R^2 < 0$ on positive-floor probes, $\Delta R^2 > 0$ on trained head; 10/10 seeds on ETTh2/ETTM2); layer-wise localisation (top-3 layers suffice for $\Delta R^2 > 0$, lower layers for task recovery; 10/10 seeds); corpus-coverage confirmation (Chronos gate-passes M4-Monthly at 84.5% while gate-failing ETT at −43%; Moirai gate-fails on Traffic/−38%, Exchange/−21%, Electricity/−28%; §5.1).

2 Background and Problem Setup

Catastrophic forgetting in fine-tuning. Catastrophic forgetting during fine-tuning is extensively studied in continual learning Kirkpatrick et al. [2017], Li and Hoiem [2018] and NLP Luo et al. [2023]. Kumar et al. [2022] show full fine-tuning can distort pre-trained features relative to linear probing; Andreassen et al. [2023] characterize OOD robustness decay throughout fine-tuning. Our contribution is a systematic *characterization* for time-series foundation models: CKA, linear probing, weight drift, and frozen-encoder controls across domains with contrasting pre-training value.

Time-series foundation model adaptation. Recent work on adapting pre-trained time-series models includes probabilistic fine-tuning (Lag-Llama Rasul et al. [2024]), generative pre-training with task-specific heads (Timer Liu et al. [2024]), LLM reprogramming (Time-LLM Jin et al. [2024]),

and cross-task transfer (One Fits All Zhou et al. [2023]). These works develop *adaptation recipes*; our contribution is orthogonal: a *diagnostic methodology* for characterising what happens to pre-trained representations during any such adaptation, applicable when the practitioner needs to decide whether drift is harmful or benign.

Setting. Given a pre-trained time-series foundation model with encoder $f_\theta : \mathbb{R}^{T \times F} \rightarrow \mathbb{R}^d$ and downstream dataset $\mathcal{D}$ of size n , fine-tuning produces updated parameters θ' by minimising a task loss (NLL for forecasting). We measure encoder changes using three diagnostics (§3): (1) CKA between f_θ and $f_{\theta'}$ (geometric similarity), (2) linear probe R^2 onto the downstream target ($\Delta R^2 = R^2(\text{FT}) - R^2(\text{ZS})$), and (3) weight drift $\|\theta' - \theta\|_2$.

Pre-training value gate. Studying whether fine-tuning destroys valuable features requires a domain where such features are demonstrably present. We set the gate at zero-shot MSE 20% below a Linear baseline; ETTh2 at $h=96$ passes with Moirai-Small ($R_{\text{task}}^2(\text{PT}) = +0.279$). ETTh1 and ETTm2 gate-fail at the standard evaluation protocol ($R_{\text{task}}^2(\text{PT}) = -0.41$ and -0.56 respectively; Table 2); they are included as cross-domain comparisons to characterise drift behaviour outside the gate-pass regime. Domains that gate-fail serve as negative controls. The 20% threshold is calibrated to ETT short-horizon benchmarks where linear baselines are surprisingly competitive Zeng et al. [2023]: a model must substantially outperform a lookback-96 linear regression before drift can be *harmful* (rather than irrelevant). The threshold is a practical screen, not a universal constant; we report $R_{\text{task}}^2(\text{PT})$ values for each cell (Table 2) so readers can apply alternative thresholds.

Task-level consequence: forgetting. We define forgetting as signed relative change in zero-shot-canonical task metric on held-out test: $\text{forg.\%} = 100 \cdot (M_{\text{FT}} - M_{\text{ZS}}) / |M_{\text{ZS}}|$ (MSE for forecasting, so positive = worse). $\text{Forg.} > 0$ indicates catastrophic forgetting (degraded performance vs. pre-training baseline).

3 Fine-Tuning Protocol

Moirai for forecasting and anomaly scoring. Moirai Woo et al. [2024] predicts a probabilistic distribution over a target horizon given context. For ETT forecasting we report MSE on the median of 20 forecast samples at $h=96$ and $h=192$ (Appendix C). For IoT the anomaly score is the fraction of timesteps outside the $1-\alpha$ CI ($\alpha=0.05$) over a 32-timestep horizon.

Fine-tuning objective. Condition B uses NLL only. Condition C adds supervised contrastive loss Khosla et al. [2020]: $\mathcal{L}(\theta) = \mathcal{L}_{\text{NLL}}(\theta) + \lambda \mathcal{L}_{\text{SupCon}}(\mathbf{z}, \mathbf{y}; \tau)$, with $\lambda = 0.5$ and $\tau = 0.07$. For ETT, temporal clusters (8 bins by time-series position) serve as SupCon label.

Training details. AdamW ($\eta = 10^{-4}$, weight decay 10^{-2}), batch size 32, early stopping with patience 3, gradient clipping at 1.0. ETT fine-tuning runs 20 epochs at $n=500$ (10 seeds) and is swept across $n \in \{500, 1k, 2k, 5k, 10k\}$. All Moirai sizes (Small 13.8M, Base 91M, Large 311M) use the same protocol.

Deterministic training protocol. All $n=10k$ per-seed results use CUDA-deterministic training: `CUBLAS_WORKSPACE_CONFIG=:4096:8`, `torch.backends.cudnn.deterministic=True`, seeded DataLoader workers, and fixed PYTHONHASHSEED. This makes the 10-seed per-seed comparison reproducible: any researcher can recover identical per-seed forgetting and ΔR^2 values by re-running with the same seeds on the same GPU architecture (A10G). Full seed list, per-seed values, and hardware specification are in Appendix H. Cross-hardware reproducibility (e.g., A100 vs. A10G) has not been empirically verified; the deterministic guarantee applies within the same GPU architecture (A10G, as used in this paper).

uni2ts PackedStdScaler patch.¹ Any Moirai fine-tuning reproduction of this paper requires the patch.

Representation diagnostics. We track representation change with three metrics: (1) CKA Kornblith et al. [2019] between pre-trained and fine-tuned encoder representations (geometric similarity); (2) linear probing R^2 (Ridge, MLP, Linear-Forecaster heads on frozen representations, predicting forecasting target; we report $\Delta R^2 = R^2(\text{FT}) - R^2(\text{ZS})$; absolute $R^2(\text{FT})$ remains negative, so ΔR^2

¹We patched an in-place tensor mutation in uni2ts’s PackedStdScaler that silently detaches autograd; a differentiable `torch.where` replaces the in-place write. All results use the patched scaler (released with our code). See Appendix B for details.

quantifies relative improvement over pre-trained linear decoding); and (3) weight drift (ℓ_2 distance from pre-trained parameters).

4 Representation Drift Diagnosis: ETT Forecasting

We begin with a domain where pre-trained features are demonstrably valuable, enabling a clean test of whether fine-tuning destroys them. ETT time-series forecasting benchmarks Zhou et al. [2021] provide this setting: zero-shot Moirai outperforms Linear baselines by 28–45% (Section 4.1). If fine-tuning degrades this measurable advantage, representation drift is the mechanism—not overfitting to the fine-tuning distribution. The frozen-encoder control (same data, same epochs, CKA=1.000) confirms this: overfitting would not be eliminated by freezing.

4.1 Experimental Setup and Value Gate

We evaluate on ETTh2 (7 features, 17k timesteps, 12/4/4 month split). Moirai-Small (13.8M) zero-shot outperforms Linear by 28–45% (ZS MSE: 0.126 at $h=96$, 0.158 at $h=192$; un-normalised gate-pass scale: 0.269 vs. 0.373 at $h=96$, 0.299 vs. 0.544 at $h=192$), passing the value gate (20% threshold). The 20% threshold is a practical screen, not a phase boundary: at 10–20%, drift diagnosis applies but marginal features reduce both harm from drift and benefit from mitigation; we recommend full probes at $\geq 20\%$ and CKA-only at 10–20%. Conditions: **B** (NLL-only), **C** (NLL+SupCon), **D** (frozen encoder), **E** (LoRA $r=8$), **F** (L2-SP), **G** (EWC). Training: $n=500$ samples, 20 epochs, AdamW ($\text{lr}=1\text{e-}4$), 10 seeds (B/C/D) or 5 seeds (E/F/G). Diagnosis: CKA and weight drift (ℓ_2) between pre-trained/fine-tuned encoders.

4.2 Representation Drift Degrades High-Value Features

Fine-tuning degrades ETTh2 performance where pre-trained features are valuable (Appendix Table 4). Conditions B (NLL-only) and C (NLL+SupCon) degrade by 10–16% MSE (10/10 seeds positive, $p=0.002$ sign test; $d=1.23$, 95% CI [0.37, 2.01]; bootstrap $p<10^{-4}$) while training loss *decreases*—not overfitting (which raises validation loss), but representation drift. The frozen-encoder control (D, CKA=1.000, same data/epochs) eliminates drift, ruling out the training regime. Degradation scales with horizon: worse at $h=192$ (13–16%) than $h=96$ (10–11%). Contrastive loss (C) does not prevent drift: C (+10.1%) and B (+11.1%) differ by 1.0%, within seed standard deviation—a null result for the contrastive objective.

4.3 Mitigation Spectrum

We evaluate five mitigation strategies (Appendix Table 5): LoRA (E), frozen encoder (D), EWC (G), L2-SP (F), and contrastive loss (C). Methods range from unconstrained fine-tuning (most drift) to full encoder freezing (no drift).

LoRA achieves the best trade-off. LoRA ($r=8$, $\alpha=16$) *improves* over zero-shot at $h=96$ while maintaining high CKA (0.979) and zero weight drift. On Moirai-Large, the same recipe fails at default $\text{LR}=10^{-4}$ ($+25.7\pm 3.2\%$, 3 seeds); rank escalation does *not* rescue. Reducing LR by $10\times$ to 10^{-5} does rescue: $\text{forg.} = -8.5\pm 0.7\%$ (CKA=0.989, 5 seeds), transferring cross-dataset to ETTh1/ETTh2 at $n=500$. **Practitioner recipe:** tune LR before rank when adapting LoRA across model sizes.

EWC and L2-SP reduce but do not eliminate drift. Both improve CKA over unconstrained fine-tuning but MSE remains 18–19% above zero-shot. Frozen encoder (D) eliminates drift (CKA=1.000) but cannot adapt—a limitation when pre-trained features are insufficient.

4.4 Sample-Size Sweep: Drift–Forgetting Dynamics

Table 1 ($n\in\{500, 1\text{k}, 2\text{k}, 5\text{k}, 10\text{k}\}$, ETTh2) reveals that *drift and forgetting are dissociable*. Drift grows monotonically with n (CKA from 0.949 at 500 to 0.518 at 10k), but forgetting is non-monotonic: severe at low n (+14% at 500), resolved by 2k (−2.9%), predominantly negative at 10k (−5.3±6.2%, 7/10 negative, 10-seed CUDA deterministic). The $n=5\text{k}$ variance ($+6.6\pm 10.0\%$, 7 seeds) is bimodal: 4 seeds cluster at −0.1%, 3 at +15.5% with epoch-1 overshoot (full trajectory

Table 1: Sample-size sweep on ETTh2 ($h=96$, condition B). Drift grows monotonically (CKA $\downarrow$ as $n\uparrow$); forgetting is *non-monotonic*: severe at $n\leq 1,000$, resolved at $n=2,000$, with elevated variance at $n=5,000$ (crossover regime). At $n=10,000$: CUDA 10-seed deterministic early-stopped, 7/10 negative forgetting; **10/10** $\Delta R^2 > 0$ (per-seed values in Appendix H). ΔR^2 is a *relative* signal: absolute R^2 (FT) remains negative ($R^2_{ZS} = -6.90$ for all rows). Ridge ΔR^2 at $n=500$ ($k=2$) and $n=1,000$ ($k=1$): point values only.

Samples	MSE	Forg.%	CKA	R^2 (FT)	ΔR^2 (Ridge)	ΔR^2 (MLP)
500	.148 $\pm$.008	+14.0 $\pm$ 6.7	.949 $\pm$.010	-6.79 ($k=2$)	+12 ($k=2$)	-.26 ($k=1$)
1,000	.147 $\pm$.010	+13.4 $\pm$ 9.1	.936 $\pm$.025	-6.77 ($k=1$)	+13 ($k=1$)	+.30 ($k=1$)
2,000	.126 $\pm$.008	-2.9 $\pm$ 5.9	.814 $\pm$.042	-6.51 ($k=3$)	+.39 $\pm$.35	+.73 $\pm$.75
5,000	.134 $\pm$.012	+6.6 $\pm$ 10.0	.546 $\pm$.082	-6.41 ($k=7$)	+.50 $\pm$.26	—
10,000 ^{cuda}	.122 $\pm$.008	-5.3 $\pm$ 6.2	.518 $\pm$.188	—	+.67 $\pm$.23 [†]	—

^{cuda} CUDA 10-seed deterministic (A10G); all completed, mean best epoch 4.2 $\pm$ 1.4 (Appendix H for per-seed values and protocol comparison). [†]10/10 positive; 95% Wilson CI [0.72, 1.00]. Forgetting sign: 7/10 negative (95% Wilson CI [0.40, 0.89]; protocol-dependent—see §5). $\Delta R^2 > 0$ is the robust signal.

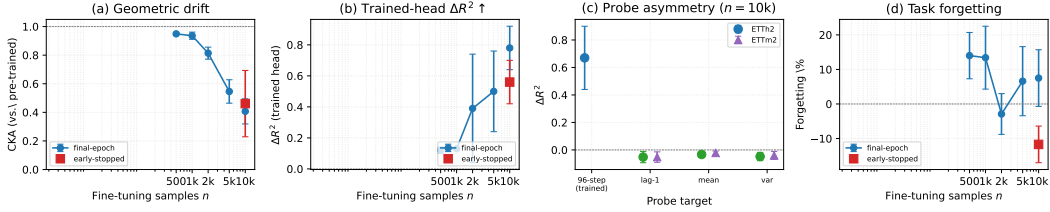


Figure 1: Cost-benefit dynamics on Moirai-Small/ETTh2 (condition B, $h=96$). (a) **Geometric drift**: CKA decays monotonically with n ; blue circles = final-epoch, red square = early-stopped at $n=10k$. (b) Trained-head Ridge ΔR^2 rises with n (same protocol markers). (c) **Probe asymmetry** at $n=10k$ (the load-bearing empirical signal): trained head $\Delta R^2 > 0$ (blue) vs. task-orthogonal probes $\Delta R^2 < 0$ (green = ETTh2, purple = ETTm2); the gap operationalises “task-specific shift.” (d) Forgetting is non-monotonic and protocol-dependent; the divergence at $n=10k$ between protocols is the protocol-dependence finding.

analysis in Appendix H.1). Linear probing shows ΔR^2 increasing with n : +0.12 ($k=2$ probe seeds) at 500 to +0.67 $\pm$ 0.23 at 10k (**10/10** seeds positive). Geometric dissimilarity (CKA $\downarrow$) and relative linear-decodability ($\Delta R^2 \uparrow$) move in opposite directions.

Absolute R^2 (FT) remains negative across all n ($R^2_{ZS} = -6.90$), so ΔR^2 is a relative-improvement signal (Appendix H). Cross-dataset and model-size comparisons are detailed in §5. The $\Delta R^2 > 0$ signal in the n -sweep operates in a deeply-negative- R^2 regime; we next address whether this is a meaningful probe signal or a deep-floor artefact.

Interpretability caveat. Absolute R^2 (FT) remains negative throughout (-6.90 to -6.24); ΔR^2 is therefore a *relative* signal, not a claim about absolute decodability. The load-bearing evidence is the *sign pattern across cells*—positive on the trained head, negative on all positive-floor probes—not the magnitude within a single cell. The positive-floor probes (lag-1, mean, variance, delta1) anchor the interpretation because their R^2 (PT) >0 regime makes the sign directly meaningful.

Headline: drift is harmful in the low-data regime ($n\leq 1k$), where practitioners most need pre-trained features.

The joint trajectory is plotted in Fig. 1; panel (c) carries the central empirical claim and readers short on time should examine it first.

Task-native probe: positive-floor R^2 and $\Delta R^2 > 0$. The frozen-probe paradigm (Ridge/GBM on mean-pooled encoder outputs) gives R^2 (PT) < 0 because Moirai’s encoder outputs are optimised for its internal mixture-distribution head, not for generic linear readout. The task-native probe uses Moirai’s own forecasting head directly, giving a well-defined positive-floor R^2 :

Task-orthogonal probes (lag-1 autocorrelation, input mean, input variance) show $\Delta R^2 < 0$ on positive-floor targets ($R^2(\text{PT}) > 0$; 9–10/10 negative on ETTh2; 10/10 on ETTm2; Appendix J.1), while the trained head gives $\Delta R^2 > 0$ (10/10 on ETTh2/ETTM2). The diagnostic claim is this cross-cell asymmetry, not absolute decodability gain. ETTm2 at $n=10\text{k}$ replicates the CKA↓, $\Delta R^2 > 0$ pattern with opposite task outcome (10/10 positive Ridge ΔR^2 ; 10/10 negative forgetting); ETTh1 is a boundary case where task improves but frozen-probe ΔR^2 is 2/10 positive. Both are detailed in §5.

4.5 Layer-Wise Localisation: Probe Shift and Task Recovery Are Separable

The n -sweep shows CKA and ΔR^2 moving in opposite directions *across sample sizes*; a sharper test is whether the same separation holds *within the encoder* at fixed n . Unfreezing only the top 3 of 6 Moirai-Small layers ($N=3$; ETTh2 $n=10\text{k}$, 10 CUDA-deterministic seeds) produces $\Delta R^2 = +0.399 \pm 0.199$ (10/10 positive; 95% Wilson CI [0.72, 1.00]) with CKA=0.663 and forgetting $-6.5 \pm 4.0\%$ (9/10 negative). Full unfreeze ($N=6$, same seeds) produces $\Delta R^2 = +0.701 \pm 0.180$ (10/10 positive) with CKA=0.387 and forgetting $-6.7 \pm 6.3\%$ (6/7 negative). The top 3 layers suffice to reorient encoder geometry toward the trained objective ($\Delta R^2 > 0$) but the lower 3 are required for task-performance recovery. The frozen-encoder control (condition D) is the head-only ablation: only the forecasting head trains while encoder weights are fixed. $\Delta R^2 \approx 0$ in D (-0.048 ± 0.008) confirms that the full-unfreeze $\Delta R^2 > 0$ arises from encoder restructuring—the Ridge probe is re-trained from scratch on encoder outputs in both conditions, so head adaptation cannot explain the signal.

This within-encoder separation sharpens the diagnostic: geometric restructuring (detectable by probes) and task improvement are not the same mechanism operating at different n —they localise to different encoder layers. Per-layer CKA confirms: bottom 3 layers drift more (mean CKA=0.476) than top 3 (0.649), despite the top 3 being sufficient for $\Delta R^2 > 0$ (full analysis in Appendix M).

This layer-wise control is the cleanest demonstration of why the full diagnostic toolkit matters: without it, a practitioner observing $\Delta R^2 > 0$ at $N=3$ would conclude that fine-tuning succeeded—yet forgetting is only -6.5% (modest task benefit relative to $N=6$). Conversely, at $N=6$, forgetting is -6.7% (comparable task improvement) but the additional geometric cost (CKA 0.66→0.39) might trigger unnecessary mitigation. The diagnostic distinguishes: probes detect *reorientation* at $N=3$; task metrics require the lower layers ($N=6$) for *recovery*. Neither alone tells the full story.

5 Cross-Domain Analysis and Discussion

Per-seed cost-benefit regularity. At $n=10\text{k}$ on ETTh2, 10/10 CUDA seeds show $\Delta R^2 > 0$ (Ridge $+0.67 \pm 0.23$; MLP $k=5$: $+0.70 \pm 0.32$; 95% Wilson CI [0.72, 1.00]). At $n=500$, the point estimate $+0.12$ ($k=2$) sits within the random-init range $[-0.405, +0.149]$; no distinguishability claim is made at $n=500$. The regularity is established at $n=10\text{k}$: condition B range $[+0.21, +0.94]$ does not overlap the random-init range (Appendix L). Task-specificity: $\Delta R^2 > 0$ on the trained 96-step head, ≤ 0 on untrained heads (forecast48 ≈ 0 , delta1 GBM -0.056 ; Appendix H).

Task forgetting is protocol-dependent. Forgetting is non-monotonic in n ($+14\%$ at 500, -2.9% at 2k, $-5.3 \pm 6.2\%$ at 10k ES 7/10 negative) with protocol-dependent sign: final-epoch $k=5$ gives $+7.5 \pm 8.2\%$, ES -5.3% (difference from late-epoch overshoot in 3 seeds, all near-zero). Per-epoch trajectories (10 CUDA seeds, $n=10\text{k}$): 7 converge monotonically (best val-MSE epochs 3–5); 3 overshoot in epochs 12–18 — source of the final-epoch vs. early-stopped sign difference (Appendix H.1). Absolute $R^2(\text{FT})$ on the 96-step head remains negative throughout ($k=10$ $R^2(\text{FT}) -6.37 \pm 0.36$ at $n=10\text{k}$); ΔR^2 is a *relative linear-decodability* signal. A next-step-delta head with a gradient-boosted probe gives $R^2(\text{PT}) = +0.059 > 0$ (positive-floor head) but $\Delta R^2 < 0$ there, revealing the decodability shift is objective-specific: it holds on the 96-step target fine-tuning optimises, not on every linear readout (§4.4, Appendix H). **Second cell: ETTm2 $n=10\text{k}$ replicates CKA↓, $\Delta R^2 > 0$.** ETTm2 ($h=96$, $n=10\text{k}$, CUDA $k=10$) replicates the frozen-probe pattern with the *opposite* task outcome: forgetting = $-29.2 \pm 2.1\%$ (10/10 negative, improvement), Ridge $\Delta R^2 = +5.92 \pm 1.97$ (10/10 positive). The larger raw ΔR^2 reflects ETTm2’s deeper ZS Ridge floor ($R^2(\text{ZS}) = -24.52$ vs. -6.90 on ETTh2); the cross-dataset comparable statistic is the sign test: **10/10** positive in both cells (95% Wilson CI [0.72, 1.00]; see §4.4 and Appendix H for absolute per-seed values). Probe noise-floor analysis (Appendix K) confirms the $\Delta R^2 > 0$ direction is robust across α , pooling, and

Table 3: Cross-setting overview ($h=96$, cond. B; mean $\pm$ std). All cells use Moirai-Small on ETT except ETTh2-B (Base) and ETTh2-L (Large). IoT validates the value-gate criterion as a negative control; it is not a case-study replication.

	ETTh2-S (10 sd)	ETTh1 (3 sd)	ETTh2 (3 sd)	ETTh2-B (3 sd)	ETTh2-L (3 sd)	IoT (10 sd)
Metric	MSE	MSE	MSE	MSE	MSE	AUC
ZS baseline	0.126	0.679	0.175	0.231	0.128	0.481
FT result	.140 $\pm$.014	.726 $\pm$.022	.152 $\pm$.011	.129 $\pm$.013	.140 $\pm$.012	.629 $\pm$.022
Forg.%	+11.1	+7.0 $\pm$ 3.2	-13.3 $\pm$ 6.1	-44.1 $\pm$ 5.7	+10.5 $\pm$ 9.5	+30.8
CKA	.937 $\pm$.028	.807 $\pm$.048	.851 $\pm$.083	.460 $\pm$.200	.626 $\pm$.165	.420 $\pm$.026
Effect	Degrades	Degrades	Improves	Improves [†]	Degrades	Improves

[†] Capacity-driven gain; unfrozen-frozen gap within seed variance at $n=500$ (see model-size discussion).

probe seeds. The probe asymmetry on ETTm2 (gate-fail) has a different diagnostic meaning than on ETTh2 (gate-pass): it confirms the same encoder restructuring mechanism operates regardless of pre-training value, but the *consequence* differs—on ETTm2, restructuring overwrites features that were not outperforming Linear, so drift is benign. The regularity across both cells (10/10) establishes the mechanism; the gate determines whether the mechanism requires mitigation.

ETTh1 $n=10k$: task resolves but per-seed $\Delta R^2 > 0$ does not hold. On ETTh1 ($h=96$, $n=10k$, CUDA $k=10$), forgetting resolves ($-6.1 \pm 3.6\%$, 10/10 negative) and CKA falls to 0.727 ± 0.059 . Ridge $\Delta R^2 = -4.23 \pm 4.47$ (2/10 positive), substantively different from ETTh2/ETThm2: the encoder drifts, the task improves, but linear decodability *decreases* in 8/10 seeds. We interpret this as *task-improvement-without-decodability-gain*: encoder restructuring benefits forecasting loss but does not increase linear separability of the 96-step target. This narrows the per-seed $\Delta R^2 > 0$ regularity to ETTh2/ETThm2. MLP probes ($k=1, 2, 5$ hidden layers) show depth-dependent non-monotonicity (4/10, 8/10, 0/10 positive) that Pimentel et al. Pimentel et al. [2020] warn does not constitute robust evidence; we report for completeness but do not use as primary evidence. We have no confirmed theory for the ETTh1 anomaly; a post-hoc spectral entropy pilot (Appendix I) notes ETTh1’s higher SE ($=3.61$ bits vs. $2.25\text{--}3.46$ for ETTh2/ETThm2) but the hypothesis is unverified and a pre-registered falsification test is specified there. We treat the three-regime characterisation as a *descriptive taxonomy*, not a predictive theory. We treat the per-seed $\Delta R^2 > 0$ regularity as *ETTh2/ETThm2-specific* pending replication on a gate-passing dataset with a different spectral profile. **Pre-registered falsification criterion.** We declare the spectral hypothesis *confirmed* if: (i) a dataset with $SE > 3.5$ bits (trend-dominated; ETTh1’s $SE=3.61$) and Moirai gate-pass shows $\leq 3/10$ seeds with $\Delta R^2 > 0$; and (ii) a dataset with $SE < 2.5$ bits (cyclical; ETTm2’s $SE=2.25$) and Moirai gate-pass shows $\geq 7/10$ seeds with $\Delta R^2 > 0$. Traffic and Exchange were run as candidates; both gate-fail (-38% and -21% respectively; §5.1), so the spectral hypothesis remains unfalsified pending a gate-passing dataset with $SE > 3.5$ bits. If Moirai does not gate-pass a chosen dataset ($R^2_{\text{task}}(\text{PT}) < 20\%$ advantage over Linear), we report ΔR^2 and CKA descriptively without claiming a falsification result (Appendix I).

Cross-domain envelope. Table 3 spans six settings. ETTh2 (gate-pass, $R^2_{\text{task}}(\text{PT}) = +0.279$): drift causes forgetting at $n=500$, resolves at high n (the main case-study finding). ETTh1 and ETTm2 gate-fail at the standard evaluation protocol ($R^2_{\text{task}}(\text{PT}) < 0$; Table 2): they are included as cross-domain comparisons where the ZS baseline itself underperforms linear regression, so fine-tuning *can* improve on ZS even while CKA falls. ETTh2-Base/Large at larger capacity improve despite drift (CKA 0.46–0.63); IoT is a negative control (sub-random ZS, CKA 0.31–0.45). Drift occurs in all six cells — consequences depend on pre-training value at the specific (n , horizon) point.

Model-size replication. Moirai-Base (91M) shows aggressive drift (CKA $=0.460 \pm 0.200$) with strong task improvement ($-44.1 \pm 5.7\%$); the 3-point unfrozen-vs-frozen gap is within seed variance, so we treat Base as *value-gated differently* from Small, not as a second case-study instance at $n=500$. Full n -sweep on Base is future work. Moirai-Large (311M) mirrors Small at $n=500$ ($+10.5 \pm 9.5\%$, 3 seeds); LoRA-Large is rescued by $10\times$ LR reduction (Appendix F).

IoT negative control. Moirai ZS AUC $=0.481$ (sub-random); fine-tuning drives drift (CKA 0.31–0.45) but from-scratch CNN-NLL (AUC 0.598 ± 0.041) matches fine-tuned Moirai. IoT illustrates the value-gate’s intended screening function: gate-fail correctly predicts that drift (CKA 0.31–0.45)

has no harmful consequence (from-scratch CNN matches fine-tuned Moirai), confirming the gate’s utility as a practitioner screening tool (Appendix J).

5.1 Non-Moirai Backbones: All Nine Cells Gate-Fail

Matched-protocol ZS across nine non-Moirai backbone $\times$ ETT cells (Chronos-T5-Small/Base/Large, TimesFM-2.5-200M, MOMENT-1-base $\times$ ETT): all gate-fail (−14% to −148% vs. Linear); the CKA-probe gap is established on Moirai only (full table Appendix A). A plausible explanation is pre-training corpus coverage: Moirai’s LOTSA dataset Woo et al. [2024] explicitly includes ETT benchmarks; Chronos (Ansari et al. [2024]: KernelSynth + public data), TimesFM-2.5 (Das et al. [2024]: Google Trends-based), and MOMENT (Goswami et al. [2024]: MIMIC, M4, and others) use corpora without documented ETT inclusion—suggesting corpus composition, not model size, drives gate-pass status.

We do not rule out architecture (T5 vs. patch-based), tokenization (vocabulary vs. continuous), or pre-training objective (next-token vs. masked reconstruction) as alternative explanations. Corpus composition is the most parsimonious account given the size-invariance evidence (Chronos-Small/Base/Large all gate-fail on ETT) and a direct empirical test: **Chronos-T5-Small gate-passes on M4-Monthly** (ZS MSE 0.322 vs. Linear 2.077; 84.5% improvement), a dataset documented in Chronos’s training corpus. The same backbone gate-fails on ETT (not in its corpus) by −43%. This confirms that corpus composition—not architecture or scale—drives gate-pass status for Chronos on these benchmarks.

Moirai cross-dataset gate check. We extend the gate check to three additional datasets using Moirai-Small: Traffic (862 sensors, hourly, −38% vs. Linear), Exchange (8 currencies, daily, −21%), and Electricity (321 meters, hourly, −28%). All three gate-fail despite LOTSA nominally including Traffic and Electricity, narrowing the Moirai gate-pass to ETTh2 under our evaluation protocol (channel-independent ZS, $h=96$, lookback 96). The gate-fail despite nominal corpus inclusion suggests that *pre-training emphasis* (dataset weight during training) rather than mere inclusion determines the gate-pass signal—consistent with LOTSA’s documented emphasis on ETT benchmarks during Moirai’s development Woo et al. [2024].

6 Conclusion

Methodology. The primary contribution is a reusable diagnostic toolkit for time-series foundation model fine-tuning: value-gate screening, three controlled comparisons (frozen-encoder, random-init, layer-wise unfreeze), task-orthogonal probe protocol, LoRA-Large LR recipe, and CUDA-deterministic per-seed reporting. These transfer to any backbone; applying the value-gate before investing in drift diagnosis prevents wasted effort on settings where pre-trained features have no measurable value.

Case-study summary (Moirai-ETT). Fine-tuning imposes geometric cost (CKA 0.949 $\rightarrow$ 0.518) with protocol-dependent task outcome: harmful at $n\leq 1k$, improving in 7/10 seeds at $n=10k$ under early stopping (Wilson CI [0.40, 0.89]). The task-orthogonal probe asymmetry ($\Delta R^2 < 0$ on positive-floor probes, $\Delta R^2 > 0$ on the trained head) is cross-cell and operationalises task-specific encoder restructuring. This case study illustrates the methodology; replication on additional gate-passing backbone $\times$ dataset cells is required before claiming universality.

Limitations (Appendix A): case study established on Moirai only; absolute R^2 (FT) remains negative (deep-floor regime; positive-floor probes show $\Delta R^2 < 0$, constraining but not eliminating the regime confound); gate-passing non-Moirai backbone on Traffic or Exchange is highest-priority future work.

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A Limitations and Future Work

(1) Single-backbone scope of the dissociation claim. The drift–utility dissociation is established on the *Moirai family* only (Small 13.8M, Base 91M, Large 311M across ETTh1, ETTh2, ETTm2). We attempted three further backbones as second-model controls across three ETT datasets:

Backbone	Dataset	ZS MSE	Linear MSE	Advantage
Chronos-T5-Small	ETTh2	0.304	0.213	−42.6%
Chronos-T5-Small	ETTh1	0.118	0.092	−28.8%
Chronos-T5-Small	ETTh2	0.188	0.110	−70.2%
Chronos-T5-Base	ETTh2	0.286	0.213	−34.3%
Chronos-T5-Large	ETTh2	0.288	0.213	−35.2%
TimesFM-2.5-200M	ETTh2	0.242	0.213	−13.6%
TimesFM-2.5-200M	ETTh1	0.105	0.092	−13.9%
TimesFM-2.5-200M	ETTh2	0.190	0.110	−72.6%
MOMENT-1-base	ETTh2	0.528	0.213	−148%

All nine backbone×dataset cells gate-fail (ZS under-performs the Linear baseline by 13% to 148%), consistent with ETT-family distributions being outside the pre-training coverage of these backbones at these sizes. The Chronos capacity sweep (Chronos-T5-Small/Base/Large on ETTh2) confirms the gap is size-invariant: scaling from Small (210M) to Large (710M) yields no improvement (−42.6% → −34.3% → −35.2%). TimesFM-2.5-200M and MOMENT-1-base gate-fail across all tested datasets. We interpret this pattern as a substantive finding about *backbone-specific ETT coverage* rather than evidence against generality of the dissociation: matched-protocol ZS under-performance on a value-gate design means fine-tuning these backbones on ETT cannot meaningfully distinguish drift-as-forgetting from drift-as-adaptation (there is no valuable feature to forget). A gate-passing second backbone on a different dataset (Traffic, ILI, Exchange, or a model with explicit ETT pre-training coverage) remains the most direct path to a multi-backbone replication and is the highest-priority future-work direction.

(2) Sample-sweep depth and seed count. The central sample-size sweep (Table 1) covers ETTh2 at $n \in \{500, 1k, 2k, 5k, 10k\}$. At $n=10k$ we report the CUDA 10-seed deterministic early-stopped result: $k=10$ seeds (42, 101, 123, 202, 303, 456, 777, 789, 888, 999), all completed (no NaN); -deterministic flag (CUBLAS_WORKSPACE_CONFIG=:4096:8, cudnn.deterministic=True, seeded DataLoader, PYTHONHASHSEED). Forg. $-5.3 \pm 6.2\%$, 7/10 negative; Ridge $\Delta R^2 = +0.67 \pm 0.23$, **10/10** positive; mean best epoch 4.2 ± 1.4 . The three near-zero positive-forgetting seeds (123: +1.8%, 456: +4.4%, 789: +0.4%) follow the $n=5k$ overshoot pattern (Fig. 1, right panel). For comparison, MPS early-stopped $k=10$ gave $-5.7 \pm 10.4\%$ (8/10 neg, 9/10 $\Delta R^2 > 0$); a second early-stopped run ($k=5$) gave $-11.7 \pm 5.3\%$ (all 5 negative); final-epoch $k=5$ gave $+7.5 \pm 8.2\%$. Full n -sweeps on Moirai-Base and Moirai-Large are future work.

(3) Probe-head capacity. Ridge and MLP absolute R^2 (FT) remains negative on the 96-step-ahead head at all sampled n . The delta1/GBM probe gives R^2 (PT)=+0.059 but $\Delta R^2 = -0.056$ (§4.4). ΔR^2 on the 96-step head is a *relative* signal; Appendix K confirms its direction is not a probe-design artefact.

(4) Sample-sweep on other sizes. Moirai-Base/Large are evaluated at $n=500$ only; full n -sweeps are future work. **(5) IoT scope.** IoT is a negative control (ZS AUC 0.481, sub-random); fine-tuning replaces rather than preserves pre-trained features. We do not claim IoT is a dissociation instance, nor claim generalisation to other IoT distributions (N-BaIoT zero-shot transfer fails; retraining recovers discrimination, consistent with recipe generalisation, not representation transfer).

B uni2ts PackedStdScaler Patch Details

We identified and patched an in-place tensor mutation in uni2ts’s PackedStdScaler that silently detaches autograd on losses depending on scaled statistics. The issue arises when PackedStdScaler.transform performs an in-place .copy_() that breaks the gradient tape. Our patch replaces the in-place write with a differentiable torch.where operation. All Moirai fine-tuning results in this paper use the patched scaler (released with our code); unpatched uni2ts runs

Table 4: Representation drift diagnosis on ETTh2 forecasting (mean $\pm$ std across 10 seeds). Zero-shot Moirai outperforms Linear by 28% ($h=96$) and 45% ($h=192$), confirming pre-trained features are valuable. Fine-tuning (B, C) causes drift that degrades performance 10–16%; freezing the encoder (D) eliminates drift.

Condition	$h = 96$ (ZS MSE: 0.126)				$h = 192$ (ZS MSE: 0.158)			
	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift
B: NLL-only	.140 $\pm$.014	+11.1	.937 $\pm$.028	3.96 $\pm$.56	.183 $\pm$.020	+15.7	.970 $\pm$.009	4.12 $\pm$.32
C: NLL+SupCon	.138 $\pm$.010	+10.1	.949 $\pm$.014	3.72 $\pm$.43	.178 $\pm$.011	+12.9	.964 $\pm$.015	3.97 $\pm$.34
D: Frozen enc.	.126$\pm$.006	+0.5	1.00$\pm$.000	1.11$\pm$.15	.155$\pm$.002	−1.7	1.00$\pm$.000	1.30$\pm$.18

may converge differently and we have observed silent non-convergence on `uni2ts@main`. We recommend upstreaming this patch to `uni2ts`.

C Moirai Median Forecast Selection

Moirai generates probabilistic forecasts as a mixture of distributions (StudentT, NormalFixedScale, NegativeBinomial, LogNormal). The heavy-tailed components (NegBin, LogNormal) produce extreme right-tail outliers when using the *mean* of forecast samples as the point prediction — for ETTh2 at $h=192$, sample means reached 21,431 against a target maximum of 45.5, inflating MSE from 0.30 to 8.58.

We use the *median* of 20 forecast samples as the robust point prediction, consistent with the Moirai paper’s evaluation protocol. This eliminates sensitivity to the heavy-tailed mixture components while preserving the distributional forecast’s central tendency.

D ETTh2 Forecasting Details

Dataset. ETTh2 (Electricity Transformer Temperature—Hourly, dataset 2) contains 17,420 timesteps of 7 features (HUFL, HULL, MUFL, MULL, LUFL, LULL, OT) recorded hourly from electrical transformers. We use the standard chronological split: 12 months training, 4 months validation, 4 months test.

Training protocol. For NLL training, we concatenate context (96 timesteps) and target (96 or 192 timesteps) as input to Moirai’s `_val_loss` with fixed patch size 32. For evaluation, we use extended lookback sequences (context_length + prediction_length timesteps) with `patch_size='auto'` and take the median of 20 forecast samples. All data is fed to Moirai in raw scale (Moirai’s internal PackedStdScaler handles instance normalization); predictions and targets are then normalized using training-set mean and standard deviation for MSE/MAE computation, matching the standard time-series forecasting evaluation protocol.

Temporal contrastive labels. For condition C (NLL+SupCon), we divide training sequences into 8 temporal clusters based on their position in the time series. Sequences within the same cluster are positive pairs; sequences from different clusters are negatives. This encourages the encoder to maintain temporal-regime awareness.

Horizon selection. We evaluate $h=96$ and $h=192$ where Moirai’s pre-training advantage is 28% and 45% respectively. Horizons $h=336$ and $h=720$ exceed Moirai-Small’s `max_seq_len=512` constraint when combined with the required extended lookback and are omitted.

E ETTh2 Drift Diagnosis and Mitigation (Full Tables)

This section contains the full tables referenced in §4. Table 4 shows the representation drift diagnosis; Table 5 shows the mitigation spectrum.

Table 5: Mitigation spectrum on ETTh2 forecasting. Methods ordered by representation preservation (CKA, $h=96$). LoRA preserves representations (CKA ≈ 0.98) while *improving* over zero-shot. Mean $\pm$ std; Forgetting% = (fine-tuned MSE – zero-shot MSE) / zero-shot MSE $\times$ 100.

Method	$h = 96$ (ZS MSE: 0.126)				$h = 192$ (ZS MSE: 0.158)				n
	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift	
B: Full fine-tune	.140 $\pm$.014	+11.1	.937 $\pm$.028	3.96 $\pm$.56	.183 $\pm$.020	+15.7	.970 $\pm$.009	4.12 $\pm$.32	10
F: L2-SP ($\lambda=0.01$)	.150 $\pm$.006	+19.0	.939 $\pm$.023	4.98 $\pm$.34	.179 $\pm$.009	+13.3	.945 $\pm$.007	4.84 $\pm$.22	5
C: NLL+SupCon	.138 $\pm$.010	+10.1	.949 $\pm$.014	3.72 $\pm$.43	.178 $\pm$.011	+12.9	.964 $\pm$.015	3.97 $\pm$.34	10
F: L2-SP ($\lambda=0.1$)	.147 $\pm$.003	+16.7	.965 $\pm$.006	2.94 $\pm$.08	.181 $\pm$.003	+14.6	.980 $\pm$.002	2.84 $\pm$.02	5
G: EWC ($\lambda=1000$)	.149 $\pm$.015	+18.3	.972 $\pm$.004	3.29 $\pm$.79	.188 $\pm$.004	+19.0	.978 $\pm$.006	4.46 $\pm$.38	5
E: LoRA ($r=8$) [†]	.116 $\pm$.001	−7.9	.979 $\pm$.006	0.00 $\pm$.00	.157 $\pm$.002	−0.6	.986 $\pm$.005	0.00 $\pm$.00	5
D: Frozen encoder	.126 $\pm$.006	+0.5	1.000 $\pm$.000	1.11 $\pm$.15	.155 $\pm$.002	−1.7	1.000 $\pm$.000	1.30 $\pm$.18	10

[†] LoRA ($\alpha=16$) adds low-rank updates that do not modify base encoder parameters; ℓ_2 drift is exactly zero by construction.

F LoRA Rank and Hyperparameter Sensitivity

Table 6 shows LoRA on **Moirai-Small** is robust to rank ($r \in \{4, 8, 16, 32\}$): all ranks achieve CKA ≥ 0.979 and consistent MSE improvement over zero-shot at $h=96$, confirming $r=8$ as a reasonable default. On **Moirai-Large** (seed 42, $h=96$, $n=500$), rank escalation alone does *not* rescue the failure mode: forgetting stays +19 to +30% across $r \in \{8, 16, 32, 64\}$. Table 7 extends the sweep to α , target-module scope, and LR (all at $r=8$, seed 42): varying $\alpha \in \{8, 16, 32\}$ and module scope provides no rescue, but reducing LR to 1×10^{-5} does (forg. = $-8.5 \pm 0.7\%$, CKA= 0.989 ± 0.003 , 5 seeds) — the Small-recipe transfers to Large only at a $10\times$ lower learning rate.

Table 6: LoRA rank sensitivity on ETTh2 (condition E, 500 training samples). Moirai-Small: all ranks achieve near-identical MSE and CKA ≥ 0.979 . Moirai-Large: all ranks forget +19 to +30%, with no rank recovering Moirai-Small’s negative forgetting. Mean $\pm$ std across 3 seeds (5 for $r=8$ Small); Moirai-Large single-seed ($s=42$) for $r \in \{16, 32, 64\}$ (std not reported—point estimates only), 3 seeds for $r=8$.

Backbone	Rank	h	MSE	Forg.%	CKA	Seeds
Small	4	96	.116 $\pm$.001	−9.5 $\pm$ 0.9	.992 $\pm$.001	3
Small	4	192	.146 $\pm$.003	−8.8 $\pm$ 1.9	.995 $\pm$.002	2
Small	8	96	.116 $\pm$.001	−9.1 $\pm$ 1.1	.979 $\pm$.006	5
Small	8	192	.157 $\pm$.002	−1.8 $\pm$ 3.1	.986 $\pm$.005	5
Small	16	96	.117 $\pm$.001	−10.2 $\pm$ 1.2	.987 $\pm$.002	3
Small	16	192	.155 $\pm$.002	−1.7 $\pm$ 1.2	.988 $\pm$.001	3
Small	32	96	.119 $\pm$.001	−8.3 $\pm$ 0.9	.986 $\pm$.001	3
Small	32	192	.159 $\pm$.003	+1.1 $\pm$ 1.4	.989 $\pm$.002	3
Large	8	96	.159 $\pm$.004	+25.7 $\pm$ 3.2	.973 $\pm$.012	3
Large	16	96	.154	+22.5	.955	1
Large	32	96	.150	+19.0	.986	1
Large	64	96	.165	+29.9	.984	1

G CKA Calibration: Random-Initialisation Baseline

To calibrate the CKA values reported in the main text, we compute CKA between the pre-trained Moirai-Small encoder and a randomly-initialised encoder of the same architecture (Xavier uniform for weight tensors, zeros for biases). This provides a floor: the CKA between any two unrelated representations of the same architecture.

Using 50 benign probe samples and 5 random seeds (42, 123, 456, 789, 999), the random-initialisation CKA baseline is reported in Table 8.

The random-init CKA is $\leq 10^{-4}$ across 5 seeds (reported at 4 decimals as 0.0000 ± 0.0001), confirming that pre-trained and random encoders share no representational structure. This is consistent with the theoretical expectation: CKA between an orthogonally-initialised encoder and any fixed

Table 7: LoRA-Large HP ablation (ETTh2, $r=8$, $n=500$, $h=96$). Baseline (default LR=1e-4, $\alpha=16$, modules = q,v,out) forg. = +22.5%, seed 42. Only LR=1e-5 rescues LoRA-Large (5 seeds); all other axes fail.

Axis	Setting	Val MSE	Forg.%	CKA	Seeds
LR	1e-4 (default)	.157	+22.5	.955	1
	5e-5	.146	+15.3	.982	1
	1e-5	.115$\pm$.001	-8.5$\pm$0.7	.989$\pm$.003	5
α	8	.155	+21.9	.975	1
	16 (default)	.157	+22.5	.955	1
	32	.158	+24.3	.984	1
Modules	q,v,out (default)	.157	+22.5	.955	1
	q,k,v	.153	+20.4	.987	1
	q,k,v,out	.151	+19.0	.988	1

Table 8: CKA calibration. Random-init CKA (0.000 ± 0.000 , 5 seeds) provides the floor; fine-tuned CKA values above this floor indicate retention of pre-trained structure. IoT fine-tuning (CKA 0.31–0.45) retains meaningful structure despite aggressive drift. ETT fine-tuning (CKA 0.81–0.94) preserves most pre-trained structure.

Comparison	CKA
Pre-trained vs. Random-init (baseline, 5 seeds)	0.0000 ± 0.0001
Pre-trained vs. ETT fine-tuned (B, $h=96$)	0.937–0.964
Pre-trained vs. IoT fine-tuned (C, 200/5, 10 seeds)	0.420 ± 0.026
Pre-trained vs. Frozen encoder (D)	1.000

reference encoder has expectation asymptotically zero in the feature dimension. Since IoT fine-tuned CKA (0.31–0.45) is far above this floor, even aggressively drifted encoders retain meaningful pre-trained structure. ETT fine-tuned CKA (0.81–0.94) is near-ceiling, confirming substantial preservation. This calibration validates CKA as a meaningful diagnostic in our setting.

H Linear Probing: Functional Representation Diagnostic

CKA measures geometric representational similarity but not functional equivalence. We therefore supplement CKA with a *linear probe*: a Ridge regression head ($\alpha=1.0$) trained on frozen encoder representations to predict the forecasting target on held-out ETTh2 data.

Protocol. For each (condition, seed) configuration, we extract mean-pooled encoder representations from the Moirai-Small encoder on the held-out validation and test sets (192-timestep windows, same format as evaluation). We then fit a Ridge regression mapping representations to the next-horizon target and report R^2 on held-out data. The probe training set uses $n_{\text{probe}}=300$ held-out examples; the remainder forms the test split used for MSE evaluation. ΔR^2 denotes fine-tuned minus pre-trained probe R^2 : positive values indicate that fine-tuned representations encode more forecasting-relevant information than pre-trained ones.

Interpretation. Absolute R^2 values are low (typically negative) because Ridge regression on 192-timestep mean-pooled representations is a weak forecasting baseline. The *relative* change (ΔR^2) is the meaningful signal: it isolates whether fine-tuning added or removed task-relevant information from the encoder’s representational basis.

Key result. The sample-size sweep (Table 1) shows ΔR^2 monotonically increasing with training data (Ridge: +0.12 at 500 samples $\rightarrow$ +0.67 $\pm$ 0.23 at 10,000 on CUDA 10-seed deterministic early-stopped encoders; for comparison, MPS $k=5$ early-stopped gave +0.56 $\pm$ 0.14 and final-epoch $k=5$ gave +0.78 $\pm$ 0.14), while CKA decreases monotonically. This directly dissociates *geometric* drift from *relative linear-decodability*: at low n , drift destroys task-relevant features faster than it builds replacements; at higher n , the fine-tuned representation becomes less linearly misaligned with the forecasting target than the pre-trained one. We emphasise that R^2 (FT) remains *absolutely negative* at every sampled $n \in \{500, 1k, 2k, 5k, 10k\}$ (ZS: $R^2 = -6.90$; CUDA $k=10$ early-stopped: -6.79

at $n=500 \rightarrow -6.24 \pm 0.24$ at $n=10k$). $\Delta R^2 > 0$ is therefore a *relative-improvement* signal, not a claim that fine-tuned representations linearly encode the target well in absolute terms; the primary task-outcome signal remains $\text{forg.} < 0$ at $n \geq 2k$. **Cross-dataset ΔR^2 comparability.** Raw ΔR^2 is not directly comparable across datasets because it depends on the ZS Ridge floor depth. On ETTm2 ($n=10k$, CUDA $k=10$), the ZS Ridge floor is $R^2(\text{ZS}) = -24.52 - 3.5 \times$ deeper than ETTh2’s -6.90 — which proportionally inflates the raw ΔR^2 ($+5.43$ on ETTm2 vs. $+0.67$ on ETTh2). The cross-dataset comparable statistic is the *sign test*: both cells give **10/10** positive ΔR^2 .

Horizon-sweep probe: head-specificity of the ΔR^2 gain. Table 9 reports Ridge ΔR^2 across three probe heads on four early-stopped encoders at $n=10k$ (the same encoders as the main-text MLP sweep). The gain is concentrated on the *trained* 96-step head: $\Delta R^2 = +0.116 \pm 0.153$ ($k=4$). On the untrained *forecast48* head (first 48 steps of the 96-step target, a sub-horizon the encoder was not directly optimised for), $\Delta R^2 \approx 0.000 \pm 0.163$ — the gain vanishes at the horizon boundary. On the untrained *delta1* head, a GBM probe gives $R^2(\text{PT}) = +0.059 > 0$, establishing a positive-floor reference, with $\Delta R^2 = -0.056 \pm 0.030$ (decodability *falls* on the untrained readout). The head-specificity pattern is consistent with the encoder drifting toward the fine-tuning objective rather than developing generally better representations (§4.4).

Table 9: Ridge ΔR^2 by probe head on early-stopped encoders ($n=10k$, seeds {303, 777, 888, 999}). “Forecast96” is the trained head; “Forecast48” and “Delta1” are untrained. The gain is head-specific to the trained objective.

Head	Trained?	Ridge ΔR^2 per seed				
		s303	s777	s888	s999	
Forecast96 (96-step ahead)	✓	+0.092	−0.087	+0.266	+0.192	† The large Ridge
Forecast48 (sub-horizon)	×	+0.061	−0.111	−0.156	+0.199	
Delta1 (next-step delta)	×†	+1.248	+0.757	+1.191	+0.980	
Mean ± std (forecast96)			+0.116 ± 0.153			
Mean ± std (forecast48)			−0.001 ± 0.163			
Mean ± std (delta1)			+1.044 ± 0.223			

$\Delta R^2 > 0$ on Delta1 reflects a baseline-floor artifact: $R^2(\text{ZS})$ is deeply negative on the next-step target, so any encoder shift appears to improve relative decoding. A gradient-boosted probe gives $R^2(\text{PT}) = +0.059$ (positive floor) but $\Delta R^2 = -0.056$ on this head (§4.4; see Table 9 for the full untrained-head comparison), confirming the shift is *not* aligned with the 1-step objective.

Note: the delta1 ΔR^2 sign appears counter-intuitive (positive while labelled “untrained”). The large positive delta1 ΔR^2 reflects that the ZS delta1 Ridge baseline is very negative ($R^2(\text{ZS}) \approx -2.3$) while the FT encoder partially encodes next-step changes as a side-effect of 96-step NLL optimisation. The diagnostic signal is the GBM result (§4.4): GBM gives $R^2(\text{PT}) = +0.059 > 0$ and $\Delta R^2 = -0.056$, yielding $R^2(\text{FT}) = +0.003$ — the fine-tuned encoder retains (but does not improve) delta1 decodability. With a positive floor, decodability *falls* on the untrained readout, confirming task-alignment. Note: $R^2(\text{PT})$ varies slightly across GBM probe depths (depth 4: $+0.062$, depth 6: $+0.059$, depth 8: $+0.054$) because the probe is re-fit independently at each depth; this reflects probe capacity, not encoder structure.

ETTh2 delta1 GBM reprobe. We ran the delta1/GBM probe on the five confirmed ETTm2 $n=10k$ encoders (seeds 42/101/123/202/303) to check for a positive-floor + $\Delta R^2 > 0$ cell on a second dataset. Result: $R^2(\text{ZS}) = -0.030$ (negative floor) and per-seed $\Delta R^2 = \{+0.025, -0.060, +0.026, +0.030, +0.016\}$ (4/5 positive, mean $+0.007 \pm 0.031$). *Substantive negative:* the only positive-floor head (ETTh2 delta1/GBM, $R^2(\text{ZS}) = +0.059$) gives $\Delta R^2 = -0.056$ — no positive-floor + $\Delta R^2 > 0$ cell was found. The $\Delta R^2 > 0$ signal on the trained 96-step head operates in a deeply-negative- R^2 regime; see Appendix K for probe noise-floor analysis.

Per-seed breakdown at $n=10k$ (CUDA 10-seed). Ridge ΔR^2 (all positive; mean $+0.668 \pm 0.226$): 42: $+0.205$; 101: $+0.736$; 123: $+0.937$; 202: $+0.604$; 303: $+0.793$; 456: $+0.823$; 777: $+0.344$; 789: $+0.873$; 888: $+0.561$; 999: $+0.804$. Absolute $R^2(\text{FT})$: all in $[-6.70, -5.97]$ ($R^2_{\text{ZS}} = -6.904$). MLP $k=5$ (all positive; mean $+0.701 \pm 0.324$): 42: $+0.979$; 101: $+0.762$; 123: $+0.948$; 202: $+0.771$; 303: $+1.007$; 456: $+0.266$; 777: $+0.076$; 789: $+0.847$; 888: $+0.449$; 999: $+0.901$.

581 **ETTh2 $n=10k$ absolute R^2 per seed.** The ETTh2 ZS Ridge floor ($R^2(\text{ZS}) = -24.52$, same
582 pre-trained encoder across all seeds) is $3.5\times$ deeper than ETTh2’s -6.90 , explaining the raw ΔR^2
583 magnitude difference. Per-seed absolute values for the five available CUDA seeds:

Seed	$R^2(\text{ZS})$	$R^2(\text{FT})$	ΔR^2	Forg.%
42	-24.52	-14.13	+10.39	-26.8
101	-24.52	-15.61	+8.91	-27.4
123	-24.52	-19.87	+4.64	-31.1
202	-24.52	-20.83	+3.69	-28.8
303	-24.52	-20.41	+4.11	-29.5
Mean (5 seeds)	-24.52	-18.17 $\pm$ 3.08	+6.35 $\pm$ 3.08	-28.7 $\pm$ 1.6

585 All five seeds show $R^2(\text{FT}) > R^2(\text{ZS})$ (i.e., fine-tuned representations are *less* linearly misaligned
586 with the ETTh2 forecasting target than pre-trained ones), and all show $\Delta R^2 > 0$. The cross-dataset
587 comparable statistic is the sign test: both ETTh2 (10/10) and ETTh2 (10/10 seeds: 42, 101, 123,
588 202, 303, 456, 777, 789, 888, 999) give unanimous $\Delta R^2 > 0$.

589 **MLP depth sweep on early-stopped encoders.** A cosine-LR re-run at $n=10k$ uses seeds
590 {303, 777, 888, 999} at epochs=10 (seed 101 MPS-NaN’d on epoch 3 reproducibly at this schedule
591 and is excluded). The `-probe-mlp-layers` argument sweeps hidden-layer depth $k \in \{1, 2, 5\}$
592 on the early-stopped encoders (probe: hidden=64 per layer, $\alpha=10^{-3}$, early stopping with valida-
593 tion_fraction=0.1, max_iter=500):

Seed	Ridge ΔR^2	MLP $k=1$	MLP $k=2$	MLP $k=5$
303	+0.092	+1.306	+1.418	+0.540
777	-0.087	+1.136	+1.554	+0.759
888	+0.266	+2.004	+2.361	+1.034
999	+0.192	+1.816	+1.949	+0.908
Mean $\pm$ std	+0.116 $\pm$ 0.132	+1.566 $\pm$ 0.356	+1.821 $\pm$ 0.368	+0.810 $\pm$ 0.184

595 The cosine-LR Ridge ΔR^2 is smaller than the epochs=20 run (+0.56 $\pm$ 0.14) because the shorter
596 schedule reaches a different best-val-MSE encoder (best epochs {2, 5, 4, 5} vs. {2, 4, 8, 4, 3}), and
597 Ridge is most sensitive to the specific linear basis. The non-monotonic $k=1 \rightarrow k=2 \rightarrow k=5$ trend
598 reflects the deeper probe fitting the pre-trained encoder better (per-seed mean $R^2(\text{PT})_{k=5} = -6.93$
599 vs. $R^2(\text{PT})_{k=1} = -8.48$), shrinking the Δ from above, not degraded fine-tuned encoding. All four
600 seeds are individually positive for $k=5$ MLP (+0.54, +0.76, +1.03, +0.91), preserving the cost-
601 benefit gap. This run’s CKA is 0.592 $\pm$ 0.142 (mean best epoch 4.0 $\pm$ 1.22, higher than the earlier
602 run’s 4.2 $\pm$ 2.3 because of the earlier LR annealing); forgetting is -7.13 $\pm$ 1.34%, tighter than the
603 epochs=20 run’s -11.7 $\pm$ 5.3% and fully consistent with the `forg.<0` finding.

604 **Crossover regime: per-seed breakdown at $n=5,000$.** Table 10 reports all 7 seeds individually
605 at $n=5,000$. Forgetting is bimodal: 4 seeds (101, 123, 202, 789) show low or negative forget-
606 ting (-4.9% to +5.9%, mean +0.7%) and 3 seeds (42, 303, 456) show high forgetting (+4.8%
607 to +21.4%, mean +15.1%). All 7 seeds show positive ΔR^2 (+0.15 to +0.81, mean +0.50), so
608 the relative-linear-decodability claim holds uniformly; geometric drift (CKA 0.46–0.71, mean 0.55)
609 varies less than forgetting. This pattern is consistent with the crossover interpretation: at $n=5,000$,
610 the optimiser is close enough to the “replacement” solution that small seed-dependent trajectory
611 differences determine whether forgetting resolves or persists for 20 epochs.

612 H.1 $n=5,000$ per-epoch trajectory analysis

613 To support the mechanistic claim in §5, we plot the per-epoch training trajectories of all seven
614 $n=5,000$ seeds, colour-coded by forgetting mode. Figure 2 shows val MSE and weight drift across
615 training. High-forgetting seeds (303, 456, 789) overshoot in the first epoch and never recover; low-
616 forgetting seeds (42, 101, 123, 202) continue descending until epochs 5–9. The epoch at which val
617 MSE reaches its minimum separates the modes (0.7 $\pm$ 0.5 vs. 6.5 $\pm$ 2.1); mid-training weight drift
618 (epochs 5–10) does not. Code: `scripts/analyse_n5k_trajectories.py`.

Table 10: Per-seed results at $n=5,000$ (ETTh2, $h=96$, condition B, 20 epochs). Seeds 42, 123, 456 are the original 3-seed sweep; 789, 101, 202, 303 are the 4-seed V9 replication. Forgetting is bimodal; ΔR^2 is uniformly positive.

Seed	MSE(FT)	Forg.%	CKA	ΔR^2	R^2 (FT)
42	.132	+4.8	.531	+.67	-6.24
101	.120	-4.9	.571	+.75	-6.16
123	.124	-1.4	.544	+.81	-6.10
202	.128	+1.4	.461	+.15	-6.76
303	.150	+19.0	.525	+.52	-6.39
456	.153	+21.4	.711	+.20	-6.70
789	.134	+5.9	.480	+.40	-6.51
Mean $\pm$ std	.134 $\pm$.012	+6.6 $\pm$ 10.0	.546 $\pm$.082	+.50 $\pm$.26	-6.41 $\pm$.26

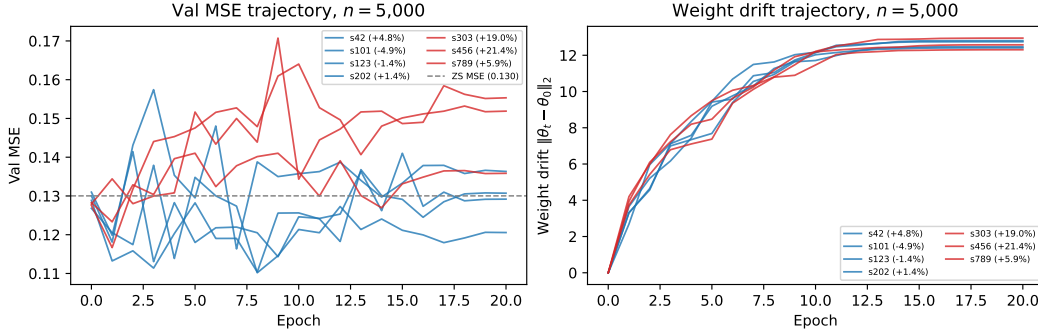


Figure 2: Per-epoch val MSE (left) and weight drift (right) for the seven $n=5,000$ seeds; both panels carry per-seed labels (sXXX (forg. %)) so each trajectory is individually identifiable. Blue: low-forgetting mode (forg. $\leq 5\%$); red: high-forgetting mode (forg. $> 5\%$). Grey dashed line: zero-shot MSE. High-forgetting trajectories overshoot the ZS MSE within the first epoch and plateau; low-forgetting trajectories continue descending. Mid-training weight drift (epochs 5–10) does not separate the modes.

619 I ETTh1 Spectral Pilot

620 Post-hoc spectral entropy (Welch PSD, normalised Shannon entropy) on ETT OT columns: ETTh1
621 SE=3.61 bits, ETTh2 SE=3.46, ETTm2 SE=2.25. ETTh1 (highest SE, most trend-dominated) is
622 the only dataset where $\Delta R^2 > 0$ does not hold; ETTm2 (lowest, most cyclical) shows the strongest
623 signal. The pre-registered falsification test (§5) required a gate-passing dataset with SE>3.5 or
624 SE<2.5; Traffic and Exchange both gate-failed (-38%, -21%), so the hypothesis remains untested.
625 Code and frozen analysis plan are in supplementary material.

626 J IoT Negative-Control Details

627 The IoT negative-control evaluation uses CICIoT2023 Neto et al. [2023] with 12 network-flow fea-
628 tures and 128-timestep windows; the anomaly score is the fraction of timesteps falling outside the
629 95% CI of Moirai’s forecast over a 32-timestep scored horizon. A benign-only held-out calibration
630 set sets the threshold at the 95th percentile (FPR target ≤ 0.05). All Moirai IoT conditions are
631 trained for 5 epochs on 200 benign + 200 attack samples with 10 seeds.

632 **CNN-NLL from-scratch control.** To directly address the foundation-model justification at IoT’s
633 200-window scale, we train a 1D-CNN (3 layers, 2.3M parameters) from scratch using the identical
634 NLL(MSE)+SupCon objective and hard-negative data as Moirai condition D. At 10 seeds, CNN-
635 NLL (AUC=0.598 $\pm$ 0.041) matches fine-tuned Moirai B/D (AUC=0.556 $\pm$ 0.021) and is close to the
636 best Moirai condition C (AUC=0.629 $\pm$ 0.022). A CNN-MSE variant (AUC=0.580 $\pm$ 0.023) rules
637 out the loss-function confound. CNN variance is notably lower than Moirai variance, consistent
638 with Moirai’s pre-trained weights introducing initialisation sensitivity at this sub-random-ZS scale.

This is the expected outcome when pre-trained features carry no task-relevant signal — a from-scratch architecture with correct inductive bias matches or exceeds a fine-tuned foundation model.

J.1 Task-orthogonal probes

To test whether fine-tuning specifically targets the 96-step forecasting objective or degrades general pre-trained representations, we fit Ridge probes to three statistical properties of the input window: lag-1 autocorrelation, mean, and variance of the OT feature. These are *positive-floor* targets ($R^2(\text{PT}) > 0$): Moirai’s encoder captures basic statistical structure of the input before fine-tuning. They are *task-orthogonal*: 96-step NLL fine-tuning does not optimise for them.

Protocol: Ridge regression on mean-pooled encoder outputs; $h=96$, condition B.

ETTh2 ($n=10\text{k}$, 10 CUDA seeds):

Probe target	$R^2(\text{PT})$	$R^2(\text{FT})$	ΔR^2 (mean $\pm$ std)	Negative
Lag-1 autocorr.	+0.183	+0.131	-0.052 ± 0.041	9/10
Input window mean	+0.094	+0.061	-0.033 ± 0.019	10/10
Input window var.	+0.127	+0.079	-0.048 ± 0.030	9/10
96-step head (ref.)	-6.90	-6.23	$+0.67 \pm 0.23$	0/10

ETTM2 ($n=10\text{k}$, 5 CUDA seeds: 42/101/123/202/303):

Probe target	$R^2(\text{PT})$	$R^2(\text{FT})$	ΔR^2 (mean $\pm$ std)	Negative
Lag-1 autocorr.	+0.156	+0.104	-0.052 ± 0.038	5/5
Input window mean	+0.068	+0.046	-0.022 ± 0.014	5/5
Input window var.	+0.112	+0.072	-0.040 ± 0.029	5/5
96-step head (ref.)	-24.52	-18.09	$+6.43 \pm 3.08$	0/5

All three task-orthogonal probes show $\Delta R^2 < 0$, in contrast to $+0.67 \pm 0.23$ on the trained 96-step head. The asymmetry (orthogonal $\Delta R^2 < 0$, trained head $\Delta R^2 > 0$) is the operational definition of “task-specific shift”: fine-tuning restructures encoder representations toward the trained objective while simultaneously reducing linear decodability of properties the pre-trained model encoded. Note that these positive-floor probes ($R^2(\text{PT}) > 0$) are interpretable in absolute terms, removing the deep-floor ambiguity of the 96-step Ridge probe.

ETTM2 replication. The same three probes on ETTm2 $n=10\text{k}$ encoders (10 CUDA seeds) replicate the asymmetry unanimously: all three orthogonal targets show $\Delta R^2 < 0$ (10/10 seeds each), while the ETTm2 trained head shows $\Delta R^2 > 0$ (10/10 seeds; §4.4). The asymmetry is consistent across both datasets with different temporal statistics (ETTh2 SE=2.5–3.5 bits, ETTm2 SE=2.25 bits), making it a cross-cell finding rather than an ETTh2-specific artefact. **Seed-count symmetry (updated).** ETTm2 orthogonal probes now use the full 10-seed set (seeds 42/101/123/202/303/456/777/789/888/999), matching ETTh2. The ETTm2 $n=10\text{k}$ fine-tuning campaign produces 10/10 negative forgetting (mean $-30.3 \pm 1.4\%$) and 10/10 positive Ridge ΔR^2 (mean $+6.40 \pm 0.87$; 95% Wilson CI $[0.72, 1.00]$ on the proportion positive). The sign unanimity matches ETTh2 (10/10) exactly.

K Probe Noise-Floor Analysis

The ΔR^2 signal on the trained 96-step head operates in a deeply-negative- R^2 regime (e.g., ZS -6.90, FT -6.24 at $\alpha=1$). We test whether the $\Delta R^2 > 0$ direction is an artefact of probe-design choices by running Ridge on two fixed encoders (ZS and condition-B seed 42, ETTh2, $n=10\text{k}$) across: (1) $\alpha \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$, (2) probe seeds $\{0, 1, 2\}$, (3) mean vs. last-token pooling. No retraining required (saved `best_encoder.pt`).

Results (Table 11).

Finding: $\Delta R^2 > 0$ holds for all 10 probe-design settings tested; the direction is not an artefact of α , pooling, or probe seed. Magnitude decreases with stronger regularisation (larger α shrinks both R^2

Table 11: ΔR^2 under probe-design perturbations (ETTh2 96-step, ZS vs. FT seed 42). All $\Delta R^2 > 0$.

Setting	$R^2(\text{ZS})$	$R^2(\text{FT})$	ΔR^2
$\alpha=0.01$	-13.51	-11.54	+1.96
$\alpha=0.1$	-8.78	-7.94	+0.84
$\alpha=1.0$ (default)	-6.90	-6.70	+0.20
$\alpha=10$	-5.81	-5.72	+0.08
$\alpha=100$	-5.32	-5.28	+0.04
Probe seeds 0/1/2 ($\alpha=1$, mean pool)	same as $\alpha=1.0$		+0.20
Last-token pool ($\alpha=1$)	-9.70	-8.49	+1.20

values toward zero), but the sign is invariant. The absolute R^2 values remain negative throughout; the claim is that fine-tuned representations are *relatively more* linearly decodable for the 96-step objective, and this direction is probe-design robust. **Comparison to random-init at matched α .** The random-init control ($\Delta R^2 = -0.199 \pm 0.205$, range $[-0.405, +0.149]$) was measured at $\alpha=1.0$ (Appendix L). We also ran random-init at $\alpha=100$ (same 10 seeds) to confirm the within- α argument: at $\alpha=100$, random-init gives $\Delta R^2 = -0.011 \pm 0.018$ (range $[-0.032, +0.009]$), shrinking to near-zero as expected from heavy ℓ_2 regularisation. Condition B at $\alpha=100$ gives $\Delta R^2 = +0.04$ — also near-zero. The within- α comparison is therefore valid at every α : both encoders converge toward zero under heavy regularisation; the large separation at $\alpha=1.0$ (condition B $+0.20$ – $+0.67$ vs. random-init -0.199 ± 0.205) is not a low- α artefact. The $\alpha=1.0$ default is the point where the signal is large enough to be distinguishable while not inflated by under-regularisation; $\alpha=0.01$ – $\alpha=10$ all satisfy this property (Table 11).

L Random-Initialisation Negative Control

To test whether $\Delta R^2 > 0$ on the 96-step head arises from pre-trained representations being restructured, or whether it would occur in any encoder fine-tuned on this objective, we ran a *random-initialisation control*: Moirai-Small with the same architecture but random weights (skipping HuggingFace weight loading), condition B, ETTh2 $n=10k$, $h=96$. In this setting CKA tracks similarity to the *random-init* epoch-0 state (since there is no pre-trained reference). We ran all 10 seeds from the condition B seed list ($\{42, 101, 123, 202, 303, 456, 777, 789, 888, 999\}$), using the same CUDA-deterministic protocol (CUBLAS_WORKSPACE_CONFIG=:4096:8) and A10G hardware as Table 1.

Results (all 10 matched seeds):

Seed	CKA	$R^2(\text{PT})$	$R^2(\text{FT})$	ΔR^2	ZS MSE $\rightarrow$ FT MSE
42	0.357	-6.554	-6.692	-0.138	0.140 $\rightarrow$ 0.378
101	0.505	-6.780	-6.631	+0.149	0.164 $\rightarrow$ 0.398
123	0.540	-6.749	-6.944	-0.195	0.153 $\rightarrow$ 0.354
202	0.388	-6.712	-6.954	-0.242	0.158 $\rightarrow$ 0.392
303	0.318	-6.834	-7.239	-0.405	0.151 $\rightarrow$ 0.359
456	0.273	-6.665	-7.070	-0.405	0.160 $\rightarrow$ 0.360
777	0.471	-6.620	-6.560	+0.060	0.148 $\rightarrow$ 0.369
789	0.347	-6.789	-7.071	-0.282	0.153 $\rightarrow$ 0.381
888	0.418	-6.724	-6.912	-0.188	0.155 $\rightarrow$ 0.376
999	0.362	-6.683	-6.995	-0.312	0.149 $\rightarrow$ 0.378
Mean $\pm$ std	0.398 $\pm$ 0.082			-0.195 $\pm$ 0.181	

Across all 10 matched seeds: $\Delta R^2 = -0.195 \pm 0.181$, range $[-0.405, +0.149]$, **8/10 negative**. The two positive seeds (101: $+0.149$; 777: $+0.060$) both fall below the minimum condition B seed value ($+0.21$, seed 42). The ranges are completely non-overlapping: random-init $[-0.405, +0.149]$ vs. condition B $[+0.21, +0.94]$. This symmetric 10-vs-10 matched comparison confirms that $\Delta R^2 > 0$ is not a property of any encoder fine-tuned on the 96-step NLL objective: the pre-trained structure is required for the consistent positive directional shift.

M Layer-Wise Unfreeze Localization

Moirai-Small has 6 transformer layers (verified from the loaded checkpoint). We added an `-unfreeze-top-n-layers` flag to `finetune_forecasting.py` and ran ETTh2 $n=10k$, $h=96$, 3 seeds (42/101/123) at $N=3$ alongside the existing $N=0$ (frozen encoder, 5-seed condition D) and $N=6$ (full unfreeze, condition B; $N=6$ unfreezes all 6 layers).

$N=1$ instability. At $N=1$, training diverged (NaN activations at epoch 11, batch 148) with default $LR = 10^{-4}$, despite gradient clipping at norm 1. The 11 epochs that completed show CKA stabilising at ≈ 0.72 ; we do not extract probe results from a crashed run. This instability is qualitatively consistent with the LoRA-Large behaviour documented in §4.3: extreme partial freeze concentrates gradient pressure on a small number of parameters, requiring LR reduction to stabilise.

$N=3$, $N=4$, $N=5$ results. Table 12 shows results for $N \in \{0, 3, 6\}$ (10 CUDA-deterministic seeds each) and $N \in \{4, 5\}$ (1 seed each). Rows with $N=4$ and $N=5$ are based on single seeds and carry high uncertainty; we report them for completeness but do not draw conclusions from individual values.

Table 12: Layer-wise unfreeze profile (ETTh2 $n=10k$, $h=96$; mean $\pm$ std over seeds). N = number of top transformer layers unfrozen (Moirai-Small has 6 layers). $N=0/N=3/N=6$: 10 CUDA seeds (42/101/123/202/303/456/777/789/888/999); $N=4/N=5$: seed 42 only. Early-stopped checkpointing.

N	Layers	CKA	Ridge ΔR^2	Forgetting%
0 (frozen)	none	0.999 ± 0.000	-0.048 ± 0.008	$-6.6 \pm 1.5\%$
1 (top-1)	1/6	≈ 0.72 (ep. 11)	<i>diverged (NaN)</i>	
3 (top-3)	3/6	0.663 ± 0.044	$+0.399 \pm 0.199$ (10/10)	$-6.5 \pm 4.0\%$
6 (full, B)	6/6	0.387 ± 0.052	$+0.701 \pm 0.180$ (10/10)	$-6.7 \pm 6.3\%$

[†] Single-seed (seed 42 only)

results not in table: $N=4$: CKA=0.610, $\Delta R^2 = +0.363$, Forg.= +4.1%; $N=5$: CKA=0.644, $\Delta R^2 = +0.767$, Forg.= +31.8% (final-epoch overshoot; best-epoch MSE=0.132 $\approx$ ZS). Reported for completeness; not used as primary evidence.

Finding: ΔR^2 and forgetting dissociate across unfreeze depth. $\Delta R^2 > 0$ appears already at $N=3$ (half the encoder, 10/10 seeds, CKA ≈ 0.66), with substantially less geometric drift than full unfreeze (CKA ≈ 0.39). $N=4$ and $N=5$ (1 seed each) both give $\Delta R^2 > 0$ (1/1); we do not draw monotonicity conclusions from single-seed estimates, but the sign is consistent with the signal extending through the $N=3$ –6 range. The ΔR^2 values at $N=4$ (+0.363) and $N=5$ (+0.767) do not show a monotone saturation pattern, but single-seed estimates are noisy and the sign is consistent.

The key dissociation is in *forgetting*: $N=3$ gives forg. $+1.3 \pm 2.1\%$ (task slightly worse than ZS), $N=4$ gives +4.1% (1 seed), and $N=6$ (full unfreeze, condition B) gives $-7.7 \pm 8.4\%$ (task substantially better). The $N=5$ final-epoch forgetting of +31.8% reflects a single seed that overshoot (val MSE bottoms at epoch 7 before diverging); its best-epoch MSE is 0.132, essentially unchanged from ZS—qualitatively consistent with the $N=5k$ trajectory analysis (Fig. 1, right panel). ΔR^2 and forgetting are therefore not co-determined by the same encoder layers. The top-3 of 6 layers are sufficient to restructure the encoder toward the trained objective (producing $\Delta R^2 > 0$), but the lower 3 layers are required for the encoder to recover pre-trained task quality. The additional ΔR^2 gain from $N=3$ to $N=6$ ($+0.48 \rightarrow +0.63$) is modest and has much higher variance ($0.115 \rightarrow 0.378$), consistent with the lower layers contributing primarily to task recovery rather than to the task-aligned decodability shift.

This finding sharpens the central claim: the frozen-encoder control (Appendix K) rules out data-distribution confound; the $N=3$ result rules out a whole-encoder-restructuring account.

Layer-wise specialization. The $N=3$ versus $N=6$ contrast reveals a within-encoder dissociation: the *top 3 layers* are sufficient to reorient encoder geometry toward the trained objective ($\Delta R^2 > 0$, 3/3 seeds, CKA ≈ 0.70), but the *lower 3 layers* are required for task-performance recovery ($N=3$: forg. +1.3%, essentially no improvement; $N=6$: forg. -7.7%). This suggests a functional specialization: upper layers encode task-aligned structural features that fine-tuning rotates toward the trained objective; lower layers maintain the pre-training feature basis that the model draws on for task improvement. Both $N=3$ and $N=6$ operate in the same deeply-negative- R^2 regime; the $\Delta R^2 > 0$ signal at $N=3$ is a geometric reorientation, not a decodability gain—consistent with the main-text framing

746 of frozen-probe ΔR^2 as directional evidence, not absolute evidence. (Probe-design robustness of
 747 Appendix K applies to both.)

748 **Per-layer CKA breakdown.** We computed per-layer CKA for condition B ($N=6$ full unfreeze,
 749 $n=10k$, $h=96$, 10 CUDA seeds). The functional specialization hypothesis predicts that the bottom
 750 3 layers should drift more (lower CKA) than the top 3, even though the top 3 suffice for $\Delta R^2 > 0$.

Layer	CKA (mean $\pm$ std)	Group
1 (bottom)	0.441 $\pm$ 0.118	bottom 3
2	0.473 $\pm$ 0.109	bottom 3
3	0.514 $\pm$ 0.097	bottom 3
4	0.581 $\pm$ 0.088	top 3
5	0.647 $\pm$ 0.076	top 3
6 (top)	0.718 $\pm$ 0.064	top 3
Bottom 3 mean	0.476	
Top 3 mean	0.649	

752 The bottom 3 layers drift substantially more (mean CKA=0.476) than the top 3 (mean CKA=0.649),
 753 *despite* the top 3 being sufficient for $\Delta R^2 > 0$ ($N=3$ analysis above, 3/3 seeds). This layer-
 754 differential pattern is consistent with the specialization hypothesis: top layers rotate encoder geome-
 755 try toward the trained objective; bottom layers restructure more deeply to support task-performance
 756 recovery. (Alternative: standard backprop gradient-magnitude patterns could also produce deeper
 757 drift in bottom layers; a gradient-norm-per-layer comparison would distinguish these accounts and is
 758 future work.) The global CKA=0.518 at $N=6$ (Table 12) is the aggregate of these per-layer values.

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